Unit Plan: Chemical Bonding

FOR THE TEACHER

Summary

The AACT high school classroom resource library has everything you need to put together a unit plan for your classroom: lessons, activities, labs, projects, videos, simulations, and animations. We constructed a unit plan using AACT resources that is designed to teach Chemical Bonding to your students.

Grade Level

High School

NGSS Alignment

The teaching resources used in this unit plan will help prepare your students to meet the performance expectations in the following standards:

- **HS-PS1-1**: Use the periodic table as a model to predict the relative properties of elements based on the patterns of electrons in the outermost energy level of atoms.
- **HS-PS1-2**: Construct and revise an explanation for the outcome of a simple chemical reaction based on the outermost electron states of atoms, trends in the periodic table, and knowledge of the patterns of chemical properties.
- **HS-PS1-3:** Plan and conduct an investigation to gather evidence to compare the structure of substances at the bulk scale to infer the strength of electrical forces between particles.
- **HS-PS1-7:** Use mathematical representation to support the claim that atoms, and therefore mass, are conserved during a chemical reaction.
- **HS-PS2-6:** Communicate scientific and technical information about why the molecular-level structure is important in the functioning of designed materials.
- **HS-ETS1-1:** Analyze a major global challenge to specify qualitative and quantitative criteria and constraints for solutions that account for societal needs and wants.
- **HS-ETS1-3:** Evaluate a solution to a complex real-world problem based on prioritized criteria and trade-offs that account for a range of constraints, including cost, safety, reliability, and aesthetics as well as possible social, cultural, and environmental impacts.
- **Scientific and Engineering Practices**:
    - Developing and Using Models
    - Analyzing and Interpreting Data
    - Engaging in Argument from Evidence
    - Constructing Explanations and Designing Solutions
    - Using Mathematics and Computational Thinking
    - Obtaining, Evaluating, and Communicating Information
    - Planning and carrying out Investigations.

Objectives

By the end of this unit, students should be able to

- Distinguish between the locations of metal atoms versus non-metal atoms on the periodic table.
- Use electronegativity values to predict whether an ionic or covalent bond is most likely to form.
- Identify compounds as ionic, covalent, or metallic based on their chemical formula.
- Predict the number of atoms needed in a molecular formula.
- Examine ratios of atoms in compounds.
- List some properties of ionic, covalent, and metallic bonds.
- Compare and contrast the basic structure of ionic and molecular compounds.
- Determine the number of valence electrons for an atom.
- Create a Lewis dot structure for an atom, covalent compound, and ionic compound.

- Predict the charge of an ion.
- Predict the molecular shape of a covalent molecule based upon its Lewis dot structure.
- Explain why stable, neutral ionic compounds are formed from cations and anions.
- Explain why different quantities of ions combine to make different compounds.
- Explain the purposes of superscripts and subscripts in chemical formulas.
- Name and write the formulas for binary and ternary ionic compounds.
- Visualize "free-moving electrons" in metallic bonding
- Identify that different metals have different properties.
- Conceptualize the impact of one electron pair domain acting upon another, and understand how those interactions result in the molecular geometries predicted by VSEPR theory.
- Describe the implications of electron pair repulsions on molecular shape.
- Understand that the molecular shape names are descriptions of the actual shape.
- Make the correlation between geometry, nonbonding pairs and molecular shape.
- Relate the shape of a molecule and the relative electronegativity values of its constituent atoms to the polarity of the molecule.
- Explain the meaning of the following: cohesion, adhesion, surface tension, and capillary action.
- Describe the unique behaviors of water molecules, and why they are important.
- Determine the polarity of molecules
- Rank molecules in order of increasing strength of van der Waals forces, given a set of structural formulas for several compounds,
- Manipulate models to demonstrate molecular orientations giving rise to London dispersion forces, dipole-dipole forces and hydrogen bonds.
- Identify the intermolecular forces present in chemical substances.
- Recognize that physical properties are related to intermolecular forces.

Chemistry Topics

This unit supports students' understanding of

- Ionic Bonding
- Covalent Bonding
- Naming Compounds
- Molecular Formulas
- Molecular Structure
- Lewis Dot Structures
- Molecular Shapes
- VSEPR Theory
- Molecular Geometry
- Electronegativity
- Polarity
- Physical Properties
- Metallic Bonds
- Electric current
- Electrons
- Resonance
- Properties of Water
- Intermolecular Forces
- London Dispersion Forces
- Dipole-dipole Forces
- Hydrogen Bonding

Time

Teacher Preparation: See individual resources.
Lesson: 8-12 class periods, depending on class level.

Materials

- Refer to the materials list given with each individual activity.

Safety

- Refer to the safety instructions given with each individual activity.

Teacher Notes

- The activities shown below are listed in the order that they should be completed.
- The number of activities you use will depend upon the level of students you are teaching.
- The teacher notes, student handouts, and additional materials can be accessed on the page for each individual activity.
- Please note that most of these resources are AACT member benefits.

Classroom Resources:

Bonding Basics

- Help students visualize how different chemical bonds form by using the Bonding Animation to introduce the concept of bonding. Examples of ionic, covalent, and polar covalent bonds are animated, and students are given a set of compounds to predict the bonding types.

- Use the Ionic & Covalent Bonding Simulation from the September 2016 issue of *Chemistry Solutions* to allow students to investigate ionic and covalent bonding. Students interact with different combinations of atoms and are tasked with determining the type of bond and the number of atoms needed to form each. The simulation visually differentiates between the transferring of electrons when forming an ionic compound and the sharing of electrons when forming a covalent compound. Students also become familiar with the molecular formula and geometric shape, as well as the naming system for each type of bond. This simulation is unlocked and can be used by your students. It also includes a teacher guide and student activity sheet.

Covalent, Ionic & Metallic Bonding and Properties

- Students construct ionic compounds by balancing the charges on cations and ions in the activity, Constructing Ionic Compounds. This activity shows students how to form stable ionic compounds, explain why different number of cations and ions are needed to form those compounds, and use superscripts and subscripts in chemical formulas. Another option is the Ionic Bonding Puzzle which provides puzzle pieces that students use to create neutral ionic compounds. Once they have made a neutral ionic compound they can use electron dot diagrams to show the formation of the compounds. Finally they will name the ionic compounds.

- Students build models of ionic and covalent compounds with the Lego Modeling of Compounds lab. By the end of this lab, they will be able to build molecular models, examine the ratio of atoms in compounds, and compare the basic structure of ionic and covalent substances.

- Use one of our ionic bonding "bracket" activities to help students demonstrate their understanding of ionic bonding and ionic properties.
    - The activity, My Name is Bond, Ionic Bond begins with pairs of students playing a game of "Ionic Compound War" to build eight compounds. Then then transfer the compounds to a "bracket" and use their knowledge of ionic bonding, along with a solubility chart, to predict the strongest and weakest bond between four pairs of ionic substances.

    - With a similar "brackets" resource, Ionic Bonding Brackets, students apply their knowledge of ionic bond strength and its relationship to melting point and solubility. After analyzing the ionic charge and radius to predict the strongest and weakest bond between four pairs of ionic substances, they will then determine which will be the least soluble.

- The demonstration, Metallic Bonding & Magnetics can be used to show your students how electrons flow through a metal using tubes made of different metals. This demo will allow your students to visualize the "free-moving electrons" in metallic bonding, understand magnetic fields, and identify that different metals have different properties because of their electron structure.

- The activity, Isn't it Ionic uses clues and questions to help students learn how to form ionic and covalent compounds. By the end of this activity, students should be able to predict ionic charges, ionic bonds, and covalent bonds. This activity can also be used to help students solve stoichiometric problems for limiting and excess reactant calculations.

- Use the lab, Ionic vs. Covalent Compounds to allow your students to compare two visually similar substances, salt and sugar. After melting a sample of each substance and analyzing their chemical composition, students draw conclusions regarding the properties of ionic and covalent compounds.

Lewis Structures, Molecular Geometry (VSEPR) and Polarity

- The Molecular Compound lesson teaches students how to name molecular compounds and create Lewis Dot Structures using a single dice and element cards. This resource includes a set of element cards for your students to use as they work through the activity.

- Introduce molecular geometry with the VSEPR Modeling activity, which has students construct physical models of molecules and then derive the arrangement of the atoms. This guided inquiry activity allows them to conceptualize the impact of one electron pair domain acting upon another. They will also understand how those interactions result in the molecular geometries predicted by VSEPR theory. Find out more about this VSEPR Modeling Activity in a related article from the September 2017 issue of *Chemistry Solutions*.

- Students can investigate the VSEPR geometry of covalent compounds in the activity, Shapes of Molecules. They draw Lewis structures, use molecular models, and determine the geometry of covalent compounds. The following molecular shapes are covered in this activity: tetrahedral, trigonal pyramidal, trigonal planar, bent, and linear. Note this activity includes a lot of repetition so that students gain as much practice as needed to master this concept.
    - Students can further explore VSEPR theory using the activity, VSEPR with Balloons. Balloons tend to take up as much space as they can when tied together, so they look like models of central atoms in VSEPR theory, making a great metaphor for the model. This resource is an extension of the Shapes of Molecules activity and includes the shapes from that activity, as well as molecular geometries that result when central atoms do not obey the octet rule: trigonal bipyramidal, octahedral, see-saw, pyramidal, square planar, and T-shaped.

- In the activity, Making Connections Between Electronegativity, Molecular Shape, and Polarity students find the electronegativity values of a variety of elements, draw the Lewis structures of molecules made with those elements, and identify the molecular shape of each molecule. Students then determine if the molecules are polar or nonpolar based on the electronegativity values of the atoms and the shape. Finally, students use Ptable.com to find information about atoms and molecules and connect what they find to observable properties.

- Students can become familiar with the special properties of water by investigating cohesion, adhesion, surface tension, and capillary action with the activity, What Makes Water So Special? Their observations will help them define the physical properties investigate, describe the unique behaviors of water, and explain why they are important.

- The Polarity lesson plan helps students learn some valuable tips for determining if a molecule is polar or nonpolar based on its Lewis Structure, VSEPR structure, and polarity. The student activity includes a "Decision Tree" to help students work through the steps of determining if a substance is polar.

Intermolecular Forces (IMFs)

- Introduce the relationship between molecular structure and properties with the lesson, The Chemistry of Water Video. Students watch a video that is part of the American Chemical Society video series *Chemistry Basics* and answer questions as it plays. This activity will help them learn about how the shape of a molecule will determine properties such as melting and boiling point.

- Students investigate intermolecular attractive forces in a lesson plan, The Great Race: A Study of van der Waals Forces by constructing molecules and determining the forces of attraction between them: London dispersion, dipole-dipole, and hydrogen bonding. Given a set of structural formulas, they then rank the molecules in order of increasing strength of van der Waals forces.

- The Intermolecular Forces & Physical Properties demonstration allows students to observe and compare the properties of surface tension, beading, evaporation, and miscibility for water and acetone. This resource includes alignment with both the AP Chemistry Curriculum Framework and NGSS.

- If you'd prefer a lab activity, use the Physical Properties lab to lead them through an investigation of how intermolecular forces affect physical properties. This lab will help them understand what happens in the freezing and melting process and how solubility works.

- If your students are tactile learners, use one or both of these resources to help them model covalent bonding and polarity with the use of string and Styrofoam balls.
    - In the activity, *Modeling Bond Polarity*, students model the pull of electrons in a bond between two elements, demonstrating covalent bonding. In particular differentiating between polar and nonpolar bonds.
    - In a similar activity, *Modeling Molecular Polarity*, students use electronegativity values and their knowledge of covalent bonding to model the bonds in a molecule. They then use that information to help them determine the overall polarity of a molecule.

- Students can investigate London dispersion and dipole-dipole intermolecular forces with the Comparing Attractive Forces simulation. In the analysis that follows the investigation, they relate IMFs (including hydrogen bonding) to physical properties, such as boiling point and solubility. The simulation was created by the Concord Consortium for AACT using Next-Generation Molecular Workbench software.
    - Another option is the Intermolecular Forces simulation, which allows students to review the three major types of intermolecular forces – London dispersion forces, dipole-dipole interactions, and hydrogen bonding – through short video clips and accompanying text. They then answer quiz questions using the relative strengths of these forces to compare different substances given their name, formula, and Lewis structure, and put them in order based on the strength of their intermolecular forces, their boiling point, or their

vapor pressure. The simulation is designed as a five question quiz for students to use multiple times.

- Wrap up your study of IMFs with the Intermolecular Forces Review lesson that helps your students review the five types of interactions (London dispersion, dipole-induced dipoles, dipole forces, hydrogen bonding, and ionic bonding). The lesson includes a PowerPoint presentation and a student note sheet to use during the review.

Extension Activities

Connect chemistry with current events with one or both of the following activities:

- The Chemistry of Hand Sanitizer and Soap lab also shows students connections between chemistry and current events. They model the interaction between hand sanitizer particles and virus particles, as well as between soap particles and virus particles. They then apply their understanding of molecular structure and intermolecular forces to analyze their observations and behavior of the particles.
- In addition to connecting chemistry with current events, give your students extra unit conversion practice with the activity, Designing an Effective Respiratory Cloth Mask. Students use unit conversion to help compare sizes of molecules, viruses, and droplets and then use them to interpret graphical data. They then use their findings to design a cloth mask that helps protect its wearer against infection by SARS-CoV-2, the coronavirus that causes COVID-19.

Culminating Projects

Do you like to end your unit with a culminating activity? We have two projects in our Molecules & Bonding resource library.

- Using Molecular Modeling, students research a molecule selected from a teacher approved list, construct a three-dimensional model of the molecule, and present their research to the class in a 7-10 minute oral presentation.
    - If you do not have the time for students to complete a long-term project, use Properties of Common Molecular Substance instead of the Molecular Modeling project. This resource allows students to apply their knowledge of molecular polarity, shape, and intermolecular forces to explain the differences in properties between different covalent substances.

- The Evolution of Materials Science in Everyday Products project connects everyday products to chemistry and helps students understand the progression of development of common items and display their knowledge through a creative video.